CMPE2600 ICA20 – ChemPlant IO and Analog Scaling

1769-L24ER-QBFC1B has :

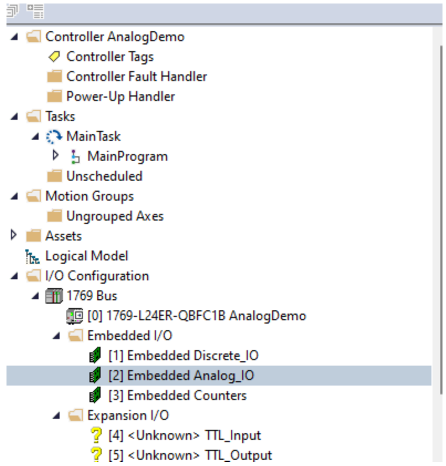
- Four embedded universal analog input points
- Two embedded analog output points
- Four embedded High-Speed Counters

The Analog IO can be configured in a few ways for different ranges of Voltages, Currents, Thermocouples, RTD's, or Resistances. See page 186 in the CompactLogix 5370 Controllers User Manual. High-Speed counters can be configured several ways as well. Since this is the case, each project needs to be setup so that each channel is configured for the proper use.

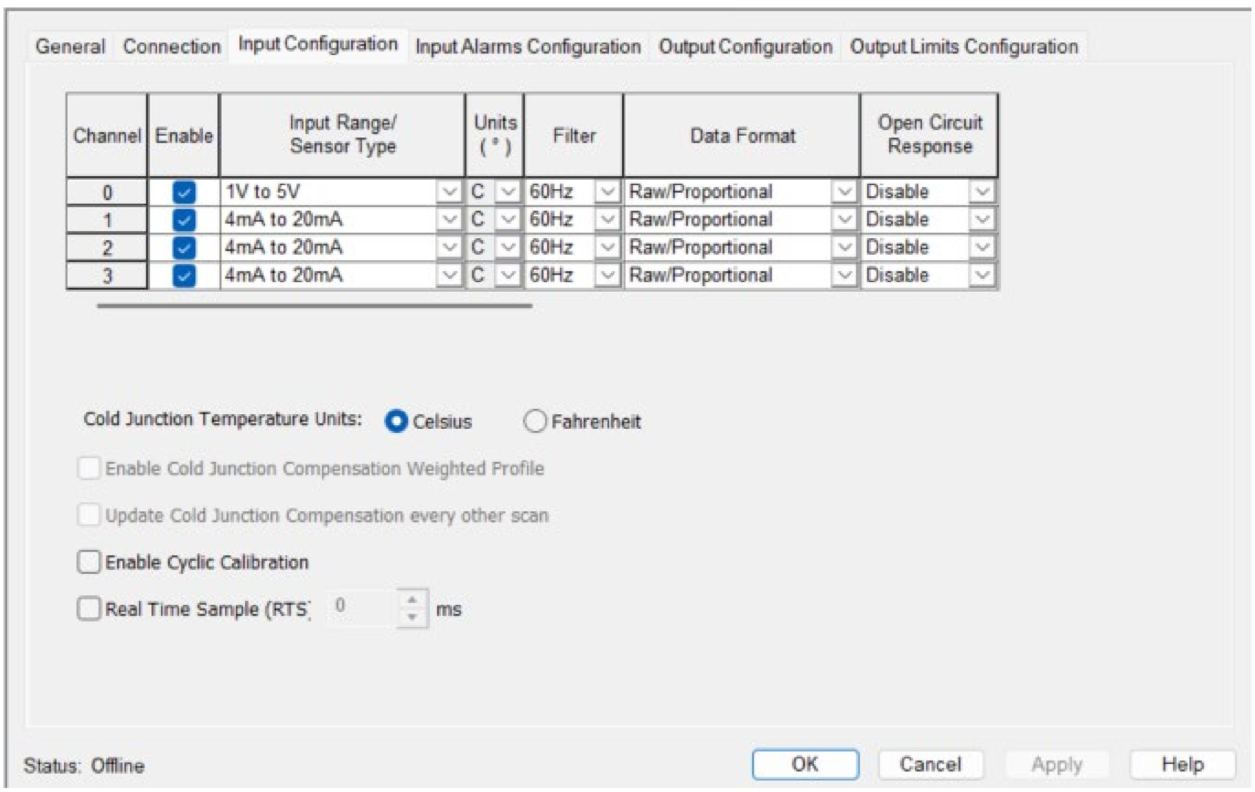
CMPE 2600 Chemplant has the following IO that is used :

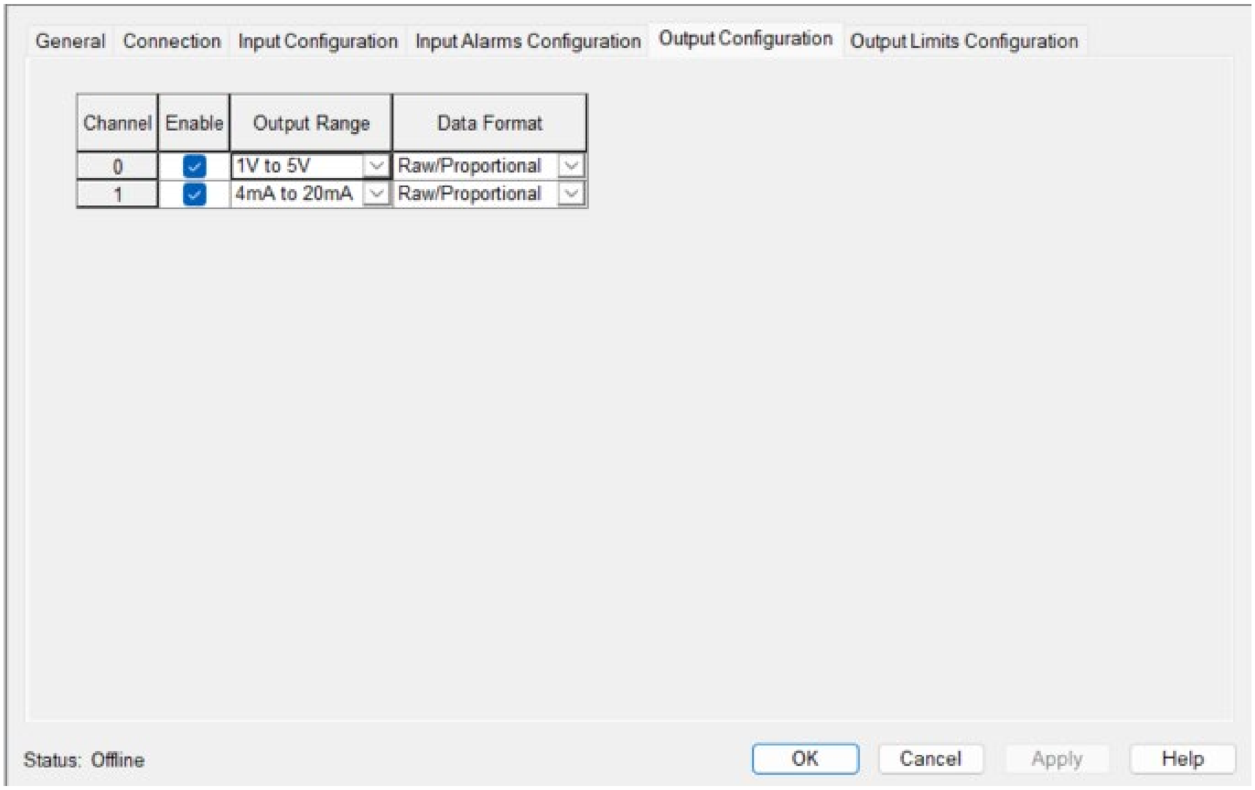
Device / IO	IO Range Raw	Digital Range	Address
Fill Valve / Output	4 – 20 mA 16 mA Range (+/- 1) Actual Input 3 – 21 mA (18 mA)	-32767 to 32767 Full Range 65535 Steps ~3641 Steps/mA (18mA)	Local:2:O.Ch1Data
Differential Pressure Sensor / Input	4 – 20 mA 16 mA Range (+/- 1) Actuel Input 3 – 21 mA (18 mA)	-32767 to 32767 Full Range 65535 Steps 3641 Steps/mA (18mA)	Local:2:I.Ch3Data
Flow Meter / Input	High-Speed Counter needed to Count pulses per Gallon.	Flow Meters used in Class range from 11500 – 15000+ pulses per Gallon	Local:3:I.Ctr0CurrentCount Local:3:O.Ctr0En Local:3:O.Ctr0SoftPreset
Pump	Digital	0 = Off 1= On	Local:1:O.Data.8
Mixer	Digital	0 = Off 1= On	Local:1:O.Data.9
Drain Valve	Digital	0 = Off 1= On	Local:1:O.Data.10

In a New Studio 5000 project, right click on the Embedded Analog_IO and select Properties.



This window should appear. Configure your Inputs and outputs as seen below





The Specific Andress for 1769-L24ER-QBFC1B Analog IO:

Analog Inputs : Local:2:I	Analog Outputs : Local:2:O
Local:2:I:Ch0Data Local:2:I:Ch1Data Local:2:I:Ch2Data Local:2:I:Ch3Data	Local:2:O:Ch0Data Local:2:O:Ch1Data

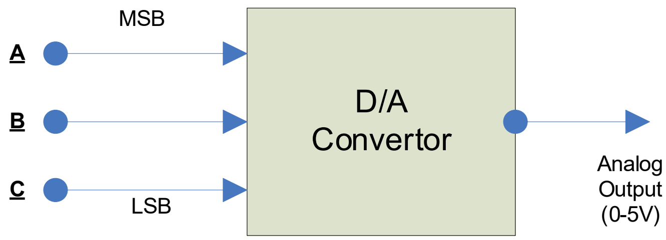
The Analog channels are word size elements made up of 16 bits, which is our resolution (16 bits = 65536).

D/A conversion needs to take place to make sense of the information. Resolution (Step size) is the smallest change that can occur in the analog output because of a change in the digital output.

Here is an example of how to calculate.

Given Analog span of 0-5V

Digital card with 3 bit resolution / channel



Step Size = Analog Span / Digital Span

= Analog Span / 2^n-1

where **n = bit resolution (3 bits)**

Substitute:

Step Size = (5-0) V / 2^3-1 = 5V / 7 = 0.714V / step

A	B	C	V out
0	0	0	0.00V
0	0	1	0.71V
0	1	0	1.42V
0	1	1	2.14V
1	0	0	2.86V
1	0	1	3.57V
1	1	0	4.28V
1	1	1	5.00V

Cannot output voltages between these values.

Embedded Analog Input Specifications

Attribute	1769-L23E-QB1B, 1769-L23E-QBFC1B, 1769-L23-QBFC1B
Inputs	4 differential or single-ended
Input range	0…10.5V 0…21 mA
Resolution	8 bits plus sign (sign is always positive).
Input impedance	Voltage: 150 kΩ nom Current: 150 Ω nom
Converter type	Successive approximation
Response speed per channel	5 ms
Rated working voltage	30V AC/30V DC
Common mode voltage	10V DC max per channel
Common mode rejection	Greater than 60 dB at 60 Hz at 10V between inputs and analog common
Normal mode rejection ratio	None
Accuracy, overall at 25 °C (77 °F)[1]	Voltage: ±0.7% full-scale, Current: ±0.6% full-scale
Accuracy, overall at 0…60 °C (32…140 °F)	Voltage: ±0.9% full-scale, Current: ±0.8% full-scale
Accuracy drift	Voltage: ±0.006% per °C, Current: ±0.006% per °C
Calibration	Not required; components provide accuracy
Non-linearity (in percent full-scale)	±0.4%
Repeatability	±0.4%
Overload at input terminals, max	Voltage: 20V continuous, 0.1 mA Current: 32 mA continuous, +5V DC
Channel diagnostics	Overrange by bit reporting
Isolation voltage	30V (continuous), basic insulation type Type tested at 500V AC for 60 s; inputs to system backplane and outputs to system backplane

(1) Includes offset, gain, non-linearity, and repeatability error terms.

https://literature.rockwellautomation.com/idc/groups/literature/documents/td/1769-td005_-en-p.pdf

Part 1

NAIT CNT CompactLogix Trainer will be used for a current source, which has a Range of ~3.65mA to ~21.35mA.

Setup

The 250-ohm resistor will me be used to convert current to voltage. Place the 250-ohm resistor across the Analog Inputs Vin0.

The Analog Input Channel 0 is setup from the previous step to take 1-5 V. The Base Tag Address Local:2:I.Ch0Data will have the digital value. The digital value will be -32767-32767. This gives 65535 total steps for the input.

The range will ~5V with 65535 steps, with .5V on each side of the range of 4V. It is same range as the chart listed above but in volts. 65535/4.5 = 14563 steps/V. 4mA per 1V gives 14563/4 = 3641 steps/mA. So at 2mA you would read -32767. Unfortunately our current source does not go to 2mA.
@4mA you should be 2 x 3641 steps from -32767.

First Calculate the digital values based on what information is provided above to fill in the Calculated Digital Value . Then using your Multimeter, measure the voltage across the 250-ohm resistor and fill out the table for Measured Digital Values . Finally Calculate the percentage.

Input Voltage from current source across 250 ohms	Calculated Digital Value	Measured Digital Values	0-100%
1.0	-25485	-25577	0
2.0	-10921	-11742	25
3.0	3643	2056	50
4.0	18204	15744	75
5.0	32767	29683	100

In Studio 5000 create an Alias Tag for the analog value. Also create a tag for Voltage and percentage. In Ladder, use the compute block to calculate the voltage and percentage into their respective tags.

Part 2

First Calculate the Digital Value that represents the input current.

Current Measurement Setup :

- From Meter connect + (mA) to the Current source +.
- From Meter connect – (COM) to the Analog Iin1+ +(Positive)
- From Current source – to Analog Iin1- -(Negative)

Use the current source adjustment and meter to setup the currents listed in the table. Then in Studio 5000, copy the value from Local:2:I:Ch1Data into the appropriate currents listed.

In Studio 5000 create an Alias Tag for the analog value. Also create a tag for Current and the percentage. In Ladder, use the compute block to calculate Current and percentage and have it show in their respective tags.

Input current from current source	Calculated Digital Value	Measured Digital Values	0-100%
3 mA		Unable to measure Current source 4-20 mA	Not part of the scale
4 mA		-29785	0
8 mA		-15015	25
12 mA		-137	50
16 mA		14497	75
20 mA		29208	100
21 mA		Unable to measure Current source 4-20 mA	Not part of the scale

<u>Part 3</u>

Main Message Display with Toggle

A main message will be something like this:

The Current is : XX.XXmA | XX%

Name	Scope	Value
Main_Message	Controller	'The Current is: 16.4132347'

Or

The Voltage is : XX.XXV | XX%

Name	Scope	Value
Main_Message	Controller	'The Voltage is: 5.04366159'

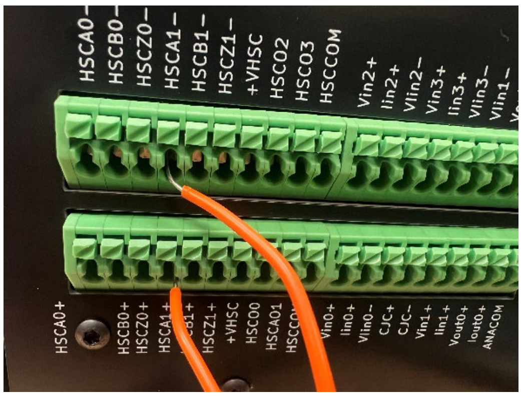
Use a Toggle Switch DI00 to switch between messages.

Using Ladder Functions such as DTOS (Dint to String), and CONCAT (String Concatenation) along with Memory tags to store partial conversion steps, as well as Static strings tags to construct a Main_Message.

Part 4

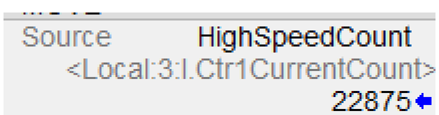
High-Speed Counter with message display

The image below shows CountEnable which is linked to DI01 Toggle switch on the PLC Trainer.   When Enabled, HSC1 will be able to count.



Use the Wave Generator at 10 – 100 Hz to connect to the HSCA1+- as shown above.

Once you have everything connected you will see the count in Local:3:I:Ctr1CurrentCount



Create a Main_Message2 for the count.   It should look something like this when completed

Add a reset for the counter using DI07. Image below shows the output you tie it to.